KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5073B | 5 Credits: 4 | Program Elective | Source-grounded

Modern Separation Techniques

 To enable the students to comprehend the principles of modern separation

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5073B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5073B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	5073B
Course title	Modern Separation Techniques
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To enable the students to comprehend the principles of modern separation
- techniques and its application in Industries.

Course outcomes

CO	Official outcome	Study level
CO1	adsorption, centrifugal separation and 12 Understanding chromatography	See official syllabus
CO2	Summarize membrane separation processes. 18 Understanding	See official syllabus
CO3	Summarize charge based separation techniques 18 Understanding Summarise liquid membranes and surfactant based	See official syllabus
CO4	12 Understanding separations	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	separation and chromatography	4	As specified
Module 2	Summarize membrane separation processes.	4	As specified
Module 3	Summarize charge based separation techniques	4	As specified
Module 4	Summarise liquid membranes and surfactant based separations	4	As specified

MODULE 1

separation and chromatography

This module is presented from the official syllabus scope for Modern Separation Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: separation and chromatography
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding and novel separation techniques Summarise the working of various adsorption

3 Understanding and novel separation techniques Summarise the working of various adsorption is an official topic in Module 1 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and novel separation techniques Summarise the working of various adsorption using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and novel separation techniques summarise the working of various adsorption should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and novel separation techniques Summarise the working of various adsorption means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding and novel separation techniques Summarise the working of various adsorption [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding equipments Study the various types of centrifugal separation

3 Understanding equipments Study the various types of centrifugal separation is an official topic in Module 1 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding equipments Study the various types of centrifugal separation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding equipments study the various types of centrifugal separation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding equipments Study the various types of centrifugal separation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding equipments Study the various types of centrifugal separation □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding techniques Illustrate the principles and types of

3 Understanding techniques Illustrate the principles and types of is an official topic in Module 1 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding techniques Illustrate the principles and types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding techniques illustrate the principles and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding techniques Illustrate the principles and types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding techniques Illustrate the principles and types of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications.

3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications. is an official topic in Module 1 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding chromatographic techniques overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. adsorption equipments – fixed or packed beds, rotary-bed adsorber, fluidized bed adsorber – applications. centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and isopycnic centrifugation. chromatography, types of chromatography- liquid chromatography, tlc, gas chromatography, column chromatography, paper chromatography, affinity chromatography, gradient chromatography- applications. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
separation and chromatography
            Summarize the difference between conventional
  M1.01                                                             3
Understanding
            and novel separation techniques
            Summarise the working of various adsorption
  M1.02                                                             3
Understanding
            equipments
            Study the various types of centrifugal separation
  M1.03                                                             3
Understanding
            techniques
            Illustrate the principles and types of
  M1.04                                                             3
Understanding
            chromatographic techniques

Contents:
Overview of conventional separation processes such as adsorption, ion exchange,
chromatography and counter current separations.
Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed
adsorber
– Applications.
Centrifugal separation – differential centrifugation, density gradient
centrifugation, ultra
centrifugation, rate zonal and Isopycnic centrifugation.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize membrane separation processes.

This module is presented from the official syllabus scope for Modern Separation Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize membrane separation processes.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding process

4 Understanding process is an official topic in Module 2 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding process using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding process [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and

Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and is an official topic in Module 2 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, study the various types of synthetic membranes 5 understanding illustrate the various membrane modules and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

5 Understanding applications Summarise the methods of membrane

5 Understanding applications Summarise the methods of membrane is an official topic in Module 2 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding applications Summarise the methods of membrane using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding applications summarise the methods of membrane should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding applications Summarise the methods of membrane means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding applications Summarise the methods of membrane □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes.

4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes. is an official topic in Module 2 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding manufacture membrane separation processes- basic principle, major areas of application, membrane materials, pore characteristics, membrane cleaning. types of synthetic membranes- microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. general methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes. □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize membrane separation processes.
            Summarise the principle of membrane separations
  M2.01                                                             4
Understanding
            process
  M2.02     Study the various types of synthetic membranes          5
Understanding
            Illustrate the various membrane modules and
  M2.03                                                             5
Understanding
            applications
            Summarise the methods of membrane
  M2.04                                                             4
Understanding
            manufacture
            Series Test – I

Contents:
Membrane separation processes-Basic principle, major areas of application, Membrane
materials, pore characteristics, membrane cleaning.
Types of synthetic membranes- Microporous, asymmetric, thin film composite, electrically
charged, inorganic membrane.
Membrane modules and application- plate and frame module, tubular module, hollow fibre
module, spiral wound module.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize charge based separation techniques

This module is presented from the official syllabus scope for Modern Separation Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize charge based separation techniques
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their

Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their is an official topic in Module 3 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the reverse osmosis process 4 understanding summarise the filtration processes and their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

5 Understanding applications

5 Understanding applications is an official topic in Module 3 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 5 Understanding applications [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using

Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using is an official topic in Module 3 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline electrodialysis and pervaporation 5 understanding outline the technique of gas separation using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model.

4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model. is an official topic in Module 3 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations,

Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding membranes reverse osmosis- concept, isotonic solutions, selection of ro membranes, high pressure and low pressure ro, osmotic pinch effect, applications. ultra filtration- basic principle, uf membranes, configuration of uf unit, types of devices in uf- funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. nanofiltration- principle, nf membranes, process limitations, industrial applications. microfiltration- cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. dialysis, electrodialysis- batch and continous electro dialysis, electrodialysis reversal (edr), permeation, per-vaporation and electrophoresis. gas separation using membranes- cross flow and counter current model. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize charge based separation techniques

  M3.01     Summarise the reverse osmosis process                   4
Understanding
            Summarise the filtration processes and their
  M3.02                                                             5
Understanding
            applications
  M3.03     Outline electrodialysis and pervaporation               5
Understanding
            Outline the technique of gas separation using
  M3.04                                                             4
Understanding
            membranes

Contents:
Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High
pressure
and low pressure RO, osmotic pinch effect, applications.
Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types
of devices
in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices,
stirred flow
modules, cross flow modules, fouling, applications.
Nanofiltration- principle, NF membranes, Process limitations, Industrial
applications.
Microfiltration-Cross flow microfiltration, dead end microfiltration,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarise liquid membranes and surfactant based separations

This module is presented from the official syllabus scope for Modern Separation Techniques. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise liquid membranes and surfactant based separations
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding using membranes Summarise the methods for foam and bubble

4 Understanding using membranes Summarise the methods for foam and bubble is an official topic in Module 4 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding using membranes Summarise the methods for foam and bubble using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding using membranes summarise the methods for foam and bubble should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding using membranes Summarise the methods for foam and bubble means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 4 Understanding using membranes Summarise the methods for foam and bubble [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding separation

2 Understanding separation is an official topic in Module 4 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding separation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding separation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding separation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 2 Understanding separation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarise froth flotation and foam fractionation

Summarise froth flotation and foam fractionation is an official topic in Module 4 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise froth flotation and foam fractionation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise froth flotation and foam fractionation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise froth flotation and foam fractionation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Summarise froth flotation and foam fractionation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation 'Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry's Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060

Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation 'Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry's Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060 is an official topic in Module 4 of Modern Separation Techniques. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation ‘Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry’s Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the concept of cryogenic fractionation 4 understanding liquid membranes- benefits, types- bulk liquid membranes, emulsion liquid membranes, thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, froth flotation, foam fractionation. cryogenic fractionation- concept and applications. text / reference t/r book title/author t1 new chemical engineering separation techniques ; schoen h. m r1 separation ‘process principles; seader j.d, and henley e.j r2 separation processes; king c.j r3 perry’s chemical engineers hand book, 6th edition; perry r.h. and. green d.w r4 handbook of separation process technology; rousseu, r.w online resources sl.no website link 1 https://nptel.ac.in/courses/103/105/103105060 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane

Aspect	What to prepare
	permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation ‘Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry’s Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation ‘Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry’s Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060 □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarise liquid membranes and surfactant based separations
            Summarise the technique of liquid separation
  M4.01                                                               4
Understanding
            using membranes
            Summarise the methods for foam and bubble
  M4.02                                                                  2
Understanding
            separation
  M4.03     Summarise froth flotation and foam fractionation           2
Understanding

  M4.04     Summarise the concept of Cryogenic fractionation           4
Understanding

            Series Test – II

Contents:
Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid
membranes,
Thin sheet supported liquid membranes, hollow fibre supported liquid membranes,
polymer
inclusion membranes.
Surfactant based separations, fundamentals of surfactants at surfaces & in
solutions, liquid
membrane permeation.
Foam and bubble separation-principle, micellar separations, external field
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Understanding and novel separation techniques Summarise the working of various adsorption. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding and novel separation techniques Summarise the working of various adsorption*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding equipments Study the various types of centrifugal separation. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding equipments Study the various types of centrifugal separation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding techniques Illustrate the principles and types of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding techniques Illustrate the principles and types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography-applications.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding chromatographic techniques Overview of conventional separation processes such as adsorption, ion exchange, chromatography and counter current separations. Adsorption equipments – fixed or packed beds, Rotary-bed adsorber, fluidized bed adsorber – Applications. Centrifugal separation – differential centrifugation, density gradient centrifugation, ultra centrifugation, rate zonal and Isopycnic centrifugation. Chromatography, Types of Chromatography- Liquid Chromatography, TLC, Gas Chromatography, Column Chromatography, Paper Chromatography, affinity chromatography, gradient chromatography- applications.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding process. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding process*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and. (3 marks)

Answer: Begin with the standard definition or identity of *Study the various types of synthetic membranes 5 Understanding Illustrate the various membrane modules and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 5 Understanding applications Summarise the methods of membrane. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding applications Summarise the methods of membrane*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes-Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding manufacture Membrane separation processes-Basic principle, major areas of application, Membrane materials, pore characteristics, membrane cleaning. Types of synthetic membranes-Microporous, asymmetric, thin film composite, electrically charged, inorganic membrane. Membrane modules and application- plate and frame module, tubular module, hollow fibre module, spiral wound module. General methods of membrane manufacture-phase inversion process, sol gel peptization, track etch method, interfacial polymerization, melt pressing, film stretching, template leaching, preparation of ion exchange membranes.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 3

Define or explain Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding applications. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using. (3 marks)

Answer: Begin with the standard definition or identity of *Outline electrodialysis and pervaporation 5 Understanding Outline the technique of gas separation using*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model.. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding membranes Reverse Osmosis- Concept, isotonic solutions, selection of RO membranes, High pressure and low pressure RO, osmotic pinch effect, applications. Ultra filtration- basic principle, UF membranes, configuration of UF unit, Types of devices in UF- Funnel, inline filters, centrifuge tube devices, diffusion devices, stirred flow modules, cross flow modules, fouling, applications. Nanofiltration- principle, NF membranes, Process limitations, Industrial applications. Microfiltration-Cross flow microfiltration, dead end microfiltration, microfiltration membranes, internal and external fouling, applications. Dialysis, Electrodialysis- Batch and continous electro dialysis, Electrodialysis reversal (EDR), Permeation, Per-vaporation and Electrophoresis. Gas separation using membranes- cross flow and counter current model.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Understanding using membranes Summarise the methods for foam and bubble. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding using membranes Summarise the methods for foam and bubble*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding separation. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding separation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□□.

Define or explain Summarise froth flotation and foam fractionation. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise froth flotation and foam fractionation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□□.

Define or explain Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation-concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation ‘Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry’s Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W

Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise the concept of Cryogenic fractionation 4 Understanding Liquid membranes- benefits, Types- Bulk liquid membranes, emulsion liquid membranes, Thin sheet supported liquid membranes, hollow fibre supported liquid membranes, polymer inclusion membranes. Surfactant based separations, fundamentals of surfactants at surfaces & in solutions, liquid membrane permeation. Foam and bubble separation-principle, micellar separations, external field induced separations, electric & magnetic field separations, Froth flotation, Foam fractionation. Cryogenic fractionation- concept and applications. Text / Reference T/R Book Title/Author T1 New Chemical Engineering Separation Techniques ; Schoen H. M R1 Separation 'Process Principles; Seader J.D, and Henley E.J R2 Separation Processes; King C.J R3 Perry's Chemical Engineers Hand book, 6th Edition; Perry R.H. and. Green D.W R4 Handbook of Separation Process Technology; Rousseu, R.W Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/103/105/103105060*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding and novel separation techniques Summarise the working of various adsorption**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding process**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Summarise the reverse osmosis process 4 Understanding Summarise the filtration processes and their**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Understanding using membranes Summarise the methods for foam and bubble**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link

- https://nptel.ac.in/courses/103/105/103105060

IN-PAGE SEARCH

Search this lesson